

# Improving the Performance of QD Photodetectors through PEAi-Based Interface Passivation

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**Abstract**—Interfacial engineering using phenethyl-ammonium iodide (PEAI) is employed to improve inverted PbS quantum dot (QD) short-wave infrared photodetectors. Placement of a PEAi layer between NiO<sub>x</sub> and the QD active layer passivates defects, suppresses recombination, and reduces low-frequency noise. Optimal treatment lowers the 1 Hz noise current density from  $3.06 \times 10^{-11}$  to  $3.04 \times 10^{-12}$  A/√Hz, yielding a specific detectivity enhancement from  $2.85 \times 10^9$  to  $1.42 \times 10^{10}$  Jones at 1150 nm. Bandwidth and temporal response are improved while maintaining operational stability, offering a CMOS-compatible pathway toward high-performance SWIR imaging.

**Keywords**—quantum dots, PbS, SWIR, photodetector, interface, PEAi

## I. INTRODUCTION

Colloidal PbS quantum dots (QDs) have emerged as a cost-effective and CMOS-compatible alternative to InGaAs for short-wave infrared (SWIR) photodetection, owing to their tunable bandgap, scalable solution processing, and compatibility with monolithic integration. Inverted device architectures, in particular, offer enhanced process versatility and are ideal for back-end-of-line (BEOL) integration with CMOS readout integrated circuits (ROICs).[1-4] However, their performance is often limited by interfacial nonidealities, especially between the QD active layer and the hole transport layer (HTL), such as NiO<sub>x</sub>, which tends to form a porous, defective surface when processed at low temperatures. These defects act as recombination centers and increase dark current noise, ultimately degrading the detectivity.

Recent efforts have highlighted the role of interfacial modification layers, *e.g.* self-assembled monolayers (SAMs), in improving interfacial quality and charge selectivity in both perovskite and PbS QD optoelectronic devices. [5-8]For instance, Chen *et al.* demonstrated that incorporating SAM-

modified thiol-treated QD-HTL and NiO<sub>x</sub> layers can drastically reduce interfacial trap density, while maintaining CMOS process compatibility.[1] Our recent work by introducing (3-Aminopropyl)triethoxysilane (APTES) at the QD active layer and electron transport layer (ETL) can also increase the detectivity of the photodetector.[9] These findings underscore the importance of molecular-level interface engineering in suppressing low-frequency noise and enhancing device stability under operational bias.

Herein, we introduce phenethyl-ammonium iodide (PEAI) as a simple and effective interfacial modifier in CMOS compatible PbS QD photodetectors with the device structure of ITO/NiO<sub>x</sub>/QD-HTL/PEAI/QD ink active layer/fullerene-ETL/BCP/Ag. The PEAi layer selectively passivates the interface between NiO<sub>x</sub> and PbS QDs, reducing nonradiative recombination and improving energy level alignment. As a result, we achieve a substantial reduction in noise current density at 1 Hz, from  $3.06 \times 10^{-11}$  to  $3.04 \times 10^{-12}$  A/√Hz, and a marked improvement in specific detectivity ( $D^*$ ) at 1150 nm, reaching  $1.42 \times 10^{10}$  Jones. These results validate the effectiveness of PEAi in advancing QD-based SWIR photodetectors and reinforce the value of interface molecular design for achieving high-performance, CMOS-integrable imaging technologies.

## II. DEVICE FABRICATION

### A. NiO<sub>x</sub> Hole Transport Layer

A 20 mg/mL aqueous dispersion of nickel oxide (NiO<sub>x</sub>) was prepared in deionized water and ultrasonicated for 30 min, followed by filtration through a 0.22 μm hydrophilic membrane. Indium tin oxide (ITO) substrates were pretreated with UV-O<sub>3</sub> for 15 min before spin-coating. The NiO<sub>x</sub> solution was spin-coated onto the ITO at 2000 rpm for 30 s and

subsequently annealed in air at 150°C for 30 min. The substrates were then cooled naturally to room temperature prior to further processing.

### B. QD-HTL and PEAi treatment

A 50 mg/mL PbS QD solution (dispersed in octane, with the first exciton absorption peak between 820-840 nm) was spin-coated onto the NiO<sub>x</sub> layer at 3000 rpm for 30 s. Subsequently, an ethanedithiol (EDT) treatment solution (0.02 vol% EDT in acetonitrile, ACN) was dropped onto the film and allowed to sit for 30 s, followed by spin-drying at 4000 rpm for 10 s. The film was then rinsed three times with pure ACN (4000 rpm, 10 s each). This procedure was repeated twice to form a ~50 nm thick PbS-EDT interlayer. The films were stored in a desiccator overnight prior to further use. PEAi was dissolved in isopropyl alcohol (IPA) at concentrations of 0, 5, 10, and 20 mg/mL and the treated samples are respectively marked as “control”, “5mg/mL”, “10 mg/mL” and “20 mg/mL”. The PEAi solution was spin-coated onto the PbS-EDT layer at 3000 rpm for 30 s.

### C. PbS QD Active Layer

The PbS active layer was prepared using a solution-phase ligand exchange (SPLE) method followed by a single-step deposition process. For the SPLE based QD ink fabrication, in a 50 mL centrifuge tube, 1.844 g PbI<sub>2</sub> and 0.64 g PbBr<sub>2</sub> were dissolved in 15 mL of N, N-dimethylformamide (DMF) under vortex agitation for 5 min until a light green, clear solution was obtained. The solution was then filtered through a 0.22 µm nylon filter. Separately, a 10 mg/mL PbS QD solution in octane (with an exciton absorption peak at 1240-1270 nm) was filtered three times with a 0.22 µm nylon filter. Then, 15 mL of the filtered PbS QD solution was slowly added to the DMF precursor solution along the inner wall of the bottle. After vortexing, a clear phase separation was observed, indicating successful ligand exchange from oleic acid to inorganic halide ligands. The upper octane phase was removed, and the remaining DMF phase was washed three times with octane following the same vortex-settle-separate procedure. The resulting DMF solution was split into two 15 mL centrifuge tubes, to which 1 mL of octane was added to prevent aggregation. The mixture was centrifuged at 7800 rpm for 3.5 min, after which the supernatant was discarded. The remaining solid was immediately dried under vacuum for 40-45 min to yield purified QD powder. The purified QD powder was redispersed in a DMF: butylamine (BTA) mixture (volume ratio 1:4) at a concentration of ~350 mg/mL. The solution was spin-coated in a two-step program: 1200 rpm for 4 s, then 2500 rpm for 40 s. The film was then annealed at 85°C for 10 min, resulting in a ~400 nm thick PbS QD active layer.

### D. Fullerene-ETL and Top Electrode

On top of the PbS active layer, a 10 nm C<sub>60</sub> layer was thermally evaporated under vacuum. Subsequently, a PCBM layer (20 mg/mL in chlorobenzene) was spin-coated at 2000

rpm for 30 s, followed by a BCP layer (0.5 mg/mL in isopropanol) at 4000 rpm for 30 s. Finally, a 100 nm thick Ag electrode was thermally evaporated under high vacuum to complete the device structure.

## III. RESULTS AND DISCUSSION

To evaluate the optoelectronic performance of the PbS QD infrared photodetectors, we systematically measured the dark current density (J-V), noise current spectra, electrochemical impedance spectroscopy (EIS), external quantum efficiency (EQE), responsivity (R), and specific detectivity (D\*) of the devices with and without PEAi interfacial modification. The corresponding results are presented in **Fig. 1**.

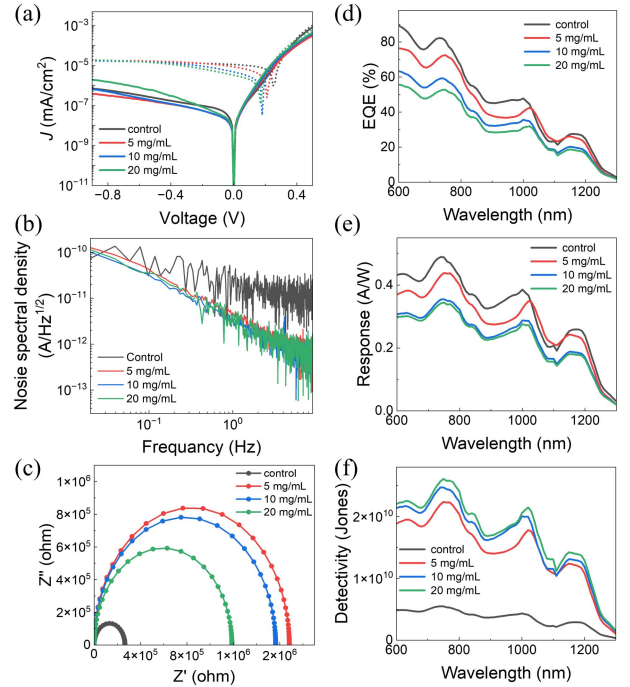


Fig. 1. (a) J-V data at the best pixel. (b) noise data at 0V bias (c) EIS data. (d) Spectra of R at bias = 0 V condition. (e) Spectra of EQE at a bias = 0 V condition. (f) D\* spectra of devices at bias = 0 V condition. (The 0/5/10/20 means that the concentration of PEAi in IPA solution is 0/5/10/20mg/ml)

As shown in **Fig. 1a**, the device treated with PEAi exhibits a slightly reduced dark current compared to the untreated device, while the photocurrent remains at a similar level across all devices. At the same time, Voc has also declined to a certain extent, indicating minimal influence of PEAi modification on the overall photocurrent generation. In contrast, the PEAi treatment results in a substantial suppression of noise current, as illustrated in **Fig. 1b**, which is the dominant factor contributing to the improvement in specific detectivity.

$$V_{oc} = \left( \frac{nk_B T}{q} \right) \ln \left( \frac{I_{ph}}{I_s} + 1 \right) \quad (1)$$

**Fig. 1f** shows that the untreated device achieves a maximum detectivity of  $2.85 \times 10^9$  Jones. After PEAi treatment, the detectivity increases significantly, reaching up to  $1.42 \times 10^{10}$  Jones for the device treated with 10 mg/mL

PEAI. This enhancement in  $D^*$  can be attributed to two main factors: (i) the significant reduction in noise current, which lowers the denominator in the detectivity expression, and (ii) a slight increase in responsivity ( $R$ ), as shown in **Fig. 1e**, which raises the numerator. Together, these effects lead to an overall improvement in the signal-to-noise performance of the photodetectors. The specific detectivity  $D^*$  is calculated using the following equation:

$$D^* = \frac{(A\Delta f)^{1/2} \times R}{i_n} \quad (2)$$

To understand the origin of performance enhancement, EIS measurements were conducted under dark conditions and fitted using the equivalent circuit according to previous work.[9] As depicted in **Fig. 1c**, the fitting results show that the series resistance ( $R_s$ ) increases from 249  $\Omega$  (pristine) to 540  $\Omega$  (5 mg/mL), 495  $\Omega$  (10 mg/mL), and 513  $\Omega$  (20 mg/mL) after PEA treatment. Which indicates that energy barriers are established by PEA treatment. Moreover, the charge transfer resistance ( $R_{ct}$ ) increases significantly from  $2.65 \times 10^5 \Omega$  to  $1.68 \times 10^6 \Omega$  (5 mg/mL),  $1.56 \times 10^6 \Omega$  (10 mg/mL), and  $1.19 \times 10^6 \Omega$  (20 mg/mL), while the capacitance ( $C$ ) remains largely unchanged. Though an increased  $R_{ct}$  can reflect an increase charge transfer barrier, it also indicates a suppressed carrier recombination at the interface, which correlates with the reduced noise current and improved  $D^*$ .

Moreover, **Fig. 1d** and **e** show that both EQE and  $R$  remain relatively stable after PEA treatment, with only a minor increase observed in responsivity. These results confirm that the enhancement in  $D^*$  mainly stems from the noise suppression effect, rather than significant gains in photocurrent generation. PEA treatment on interface between QD-HTL and QD active layer plays a critical role in passivating interfacial traps and suppressing recombination, resulting in significantly enhanced specific detectivity in inverted PbS QD photodetectors.

To further evaluate the influence of PEA treatment interface modification on the dynamic performance of the photodetectors, frequency response and temporal characteristics were systematically analyzed. As shown in **Fig. 2(a)**, the -3 dB bandwidth, which denotes the frequency at which the device response drops to 70.7% of its low-frequency value, increased significantly after PEA treatment. The pristine device exhibited a bandwidth of 56 kHz, which improved to 66 kHz, 57 kHz, and 65 kHz for devices modified with 5 mg/mL, 10 mg/mL, and 20 mg/mL PEA, respectively. The enhancement in -3 dB bandwidth indicates a faster response capability and broader detection frequency range, which is crucial for high-speed signal processing and real-time infrared imaging applications.

**Fig. 2(b)** presents the transient photoresponse measured under pulsed illumination. All PEA-modified devices demonstrated faster rise and fall times compared to the control device, confirming the improved temporal response. In which

20 mg/mL device exhibits the highest response speed with a rise time of 4.92  $\mu$ s and fall time of 5.15  $\mu$ s. High temporal resolution is essential for photodetectors used in optical communication, motion detection, and time-of-flight (ToF) imaging systems, where rapid signal transitions must be precisely captured.

In contrast, as shown in **Fig. 2(c)**, the ON-OFF switching test under periodic light modulation revealed no significant degradation in stability or repeatability across all samples. The PEA-treated devices maintained consistent ON-OFF photoresponse characteristics over multiple cycles, indicating reliable operation and long-term stability after interfacial engineering.

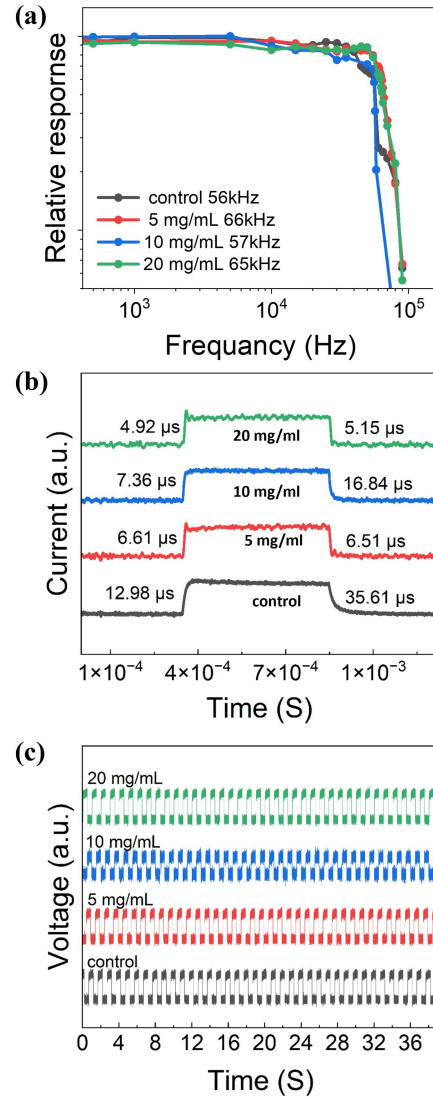


Fig. 2. (a) -3dB bandwidth. (b) Response speed (c) ON-OFF data.

#### IV. CONCLUSION

PEAI interfacial modifier was introduced between the  $\text{NiO}_x$  HTL and PbS QD active layer in inverted SWIR

photodetectors, yielding significant noise suppression and enhanced detectivity. Optimal PEA treatment (10 mg/mL) reduced noise current density at 1 Hz from  $3.06 \times 10^{-11}$  to  $3.04 \times 10^{-12}$  A/  $\sqrt{\text{Hz}}$  and improved  $D^*$  from  $2.85 \times 10^9$  to  $1.42 \times 10^{10}$  Jones. Bandwidth and temporal response were also improved without sacrificing stability, underscoring PEA's effectiveness as a simple, CMOS-compatible strategy for high-performance QD-based SWIR photodetectors.

#### ACKNOWLEDGMENT

This work was supported by National Natural Science Foundation of China (No. 12204318), Shenzhen Science and Technology Program (Nos. RCYX20221008092908030, JCYJ20241202124709012), Natural Science Foundation of Top Talent of SZTU (Nos. GDRC202340, GDRC202345).

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